PRODUCT DATASHEET · SKU CQA240229P1

8-Channel EEG Shield for Raspberry Pi

Open-source Raspberry Pi shield that turns a Raspberry Pi 4 or 5 into an 8-channel brain-computer interface. A Texas Instruments ADS1299 front end reads EEG, EMG and ECG at 24 bits, and the Pi streams, records and processes the data in real time.



8	24-bit	250 SPS	SPI0	65 × 56
Channels + REF + BIAS	TI ADS1299 front end	Default, up to 16 kSPS	40-pin GPIO header	mm · Pi HAT footprint

01 OVERVIEW

Key features

- **Plugs straight onto the Pi**: no cables between shield and host, Raspberry Pi 4 or 5 (Pi 3 also listed as compatible).
- **Research-grade front end**: TI ADS1299, 8 simultaneous 24-bit channels, programmable gain 1–24.
- **Raw access**: unfiltered 24-bit data and full control of the ADC registers from Python.
- **One-command software**: PiEEG Server with browser dashboard, WebSocket, CSV, LSL and OSC.
- **BrainFlow board**: `BoardIds.PIEEG_BOARD` (ID 56).
- **Open SDK and datasets**: MIT-licensed scripts, sample recordings and an ML example on GitHub.

Typical applications

- Teaching neuroscience and signal processing
- BCI prototypes and neurofeedback
- Alpha, blink and motor experiments
- EMG and ECG projects
- Embedded Linux and Raspberry Pi courses
- Machine-learning on biosignals

02 TECHNICAL SPECIFICATIONS

ANALOG ACQUISITION	
Channels	8 single-ended biosignal inputs
Extra inputs	Reference (SRB1, common to all channels) and bias drive
Analog front end	Texas Instruments ADS1299
Resolution	24-bit delta-sigma, all channels sampled at the same moment
Sample rate	250 SPS set by PiEEG software (ADS1299 supports 250 SPS to 16 kSPS)
Gain	1 set by PiEEG software (ADS1299 supports 1, 2, 4, 6, 8, 12, 24)
Reference voltage	4.5 V internal; 1 LSB ≈ 0.268 µV at gain 1
Signals	EEG, EMG, ECG
HOST INTERFACE	
Host	Raspberry Pi 4 or Raspberry Pi 5
Connector	40-pin GPIO socket (2 × 20, 2.54 mm)
Bus	SPI0, CE0 · mode 1 · DRDY on GPIO 26 (active low)
Data frame	27 bytes per sample: 3-byte status + 8 × 3-byte channels

For research and education only. Not a medical device. Specifications subject to change.

ELECTRODES	
Electrode type	Dry or wet Ag/AgCl
Connector	1 × 10 pin header, 2.54 mm: E1–E8, Biasout, Ref
POWER & SAFETY	
Supply	5 V through the Raspberry Pi, from a 5 V / 3 A power bank only
Indicator	Status LED, on when powered
Isolation	None. Battery operation of the whole setup is required
Classification	Research and education tool. Not a medical device
MECHANICAL	
PCB	65.14 × 56.20 mm, 2-layer FR-4, Raspberry Pi HAT footprint
Mounting	4 × Ø2.70 mm holes, 58.0 × 48.8 mm pattern (M2.5)
Enclosure	3D-printable Raspberry Pi 5 case and lid (see PE8-ENC-007)
Shipping weight	200 g (packed)
SOFTWARE	
PiEEG Server	`pieeg-server --device pieeg8` — dashboard :1617, WebSocket :1616, CSV, LSL, OSC
BrainFlow	BoardIds.PIEEG_BOARD (ID 56), 8 channels at 250 Hz
Python SDK	Scripts for Pi 4 and Pi 5, MIT licence

03 BLOCK DIAGRAM

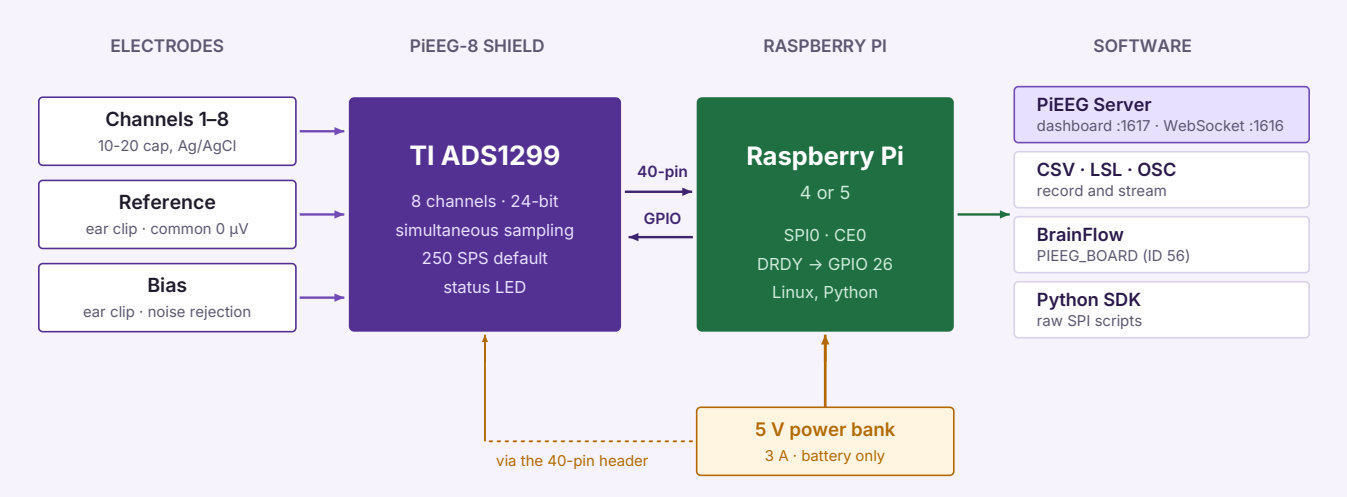


Figure 1 — Signal path. The ADS1299 digitises all channels together; the Pi reads each frame over SPI when DRDY goes low.

04 MEASURABLE SIGNALS

Signal	What it shows	Typical first test
EEG	Brain rhythms: delta, theta, alpha, beta, gamma	Close the eyes and watch alpha (8–13 Hz) rise on O1/O2
EMG	Muscle activity	Clench a fist or the jaw
ECG	Heart activity	Electrodes on the chest or wrists

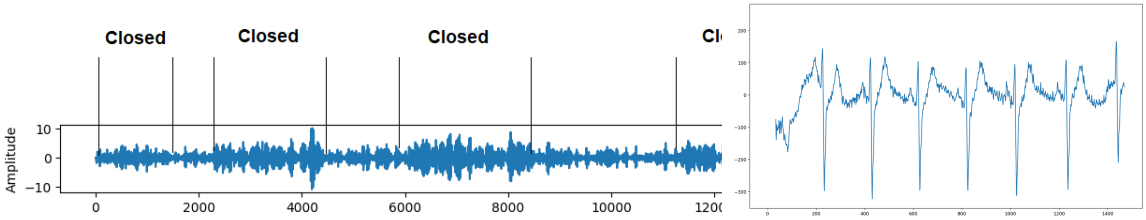


Figure 2 — Alpha rhythm appears when the eyes close.

Figure 3 — ECG recorded with PiEEG-8.

05 PIEEG FAMILY

	PiEEG-8	PiEEG-16	IronBCI-32
Channels	**8**	16	32
Host	**Raspberry Pi**	Raspberry Pi	Laptop (USB)
Converter	**1 × ADS1299**	2 × ADS1299	4 × AD7771
Sample rate	**250 SPS**	250 SPS	~500 SPS
List price	**USD 350**	USD 390	USD 480

06 ORDERING INFORMATION

Product	PiEEG-8 — 8-channel EEG shield for Raspberry Pi
SKU	CQA240229P1
Price	USD 350 (list); 5+ units USD 315 each, 10+ units USD 280 each
Purchase	Elecrow (official manufacturing partner) — elecrow.com/pieeg.html · direct store on pieeg.com coming soon
Bundle	Raspberry Pi BCI Starter Bundle: PiEEG-8 + 8-channel dry EEG cap kit + EMG/ECG kit

Included	
1 × PiEEG-8 shield	ADS1299, 40-pin socket, electrode header
Quick-start manual	Printed guide

Not included — required	
Raspberry Pi 4 or 5	microSD with Raspberry Pi OS
5 V / 3 A power bank	Battery only
Electrodes and cap	8 Ag/AgCl + 2 ear clips
Monitor, keyboard	Battery-powered, wired

WARNING Battery only.

PiEEG-8 has no isolation. The Raspberry Pi, the shield and everything cabled to them must run from a 5 V power bank while electrodes are on a person. PiEEG-8 is not a medical device.

07 RELATED DOCUMENTS

PE8-MD-002 Mechanical & Dimensional Drawing	**PE8-PCB-003** PCB & Manufacturing Specification	**PE8-PIN-004** Connector Pinout & Electrode Guide
PE8-QSG-005 Quick Start & Safety Guide	**PE8-SW-006** Data Interface & Software Guide	**PE8-ENC-007** 3D-Printed Enclosure Guide